

Do Enterprise Zones Work?

An Analysis at the Borders

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This article analyzes Enterprise Zones in Colorado to study the relationship between geographically targeted tax credits and the location of new businesses and jobs. Enterprise Zone (EZ) programs provide tax incentives for investment and job creation in economically lagging regions. While most states have EZ programs, past program evaluations have found a mixture of effectiveness. This research improves on existing literature by utilizing both establishment-level data and a border effects methodology (1) to control for unobservables that influence the self-selection of EZ regions and (2) to high-light EZ impacts across different industries. Results find that while EZ fiscal incentives have no effect on where new establishments locate in Colorado, they do increase the number of employees hired. Industry results highlight the heterogeneity of tax credit impacts within the EZ Program. Results are robust to a variety of specifications for land-use controls and in comparison to a propensity score matching model.

Keywords: *Enterprise Zones; state and local economic development; taxes and location of economic activity*

1. Introduction

State and local governments' economic development strategies commonly include tax incentives to attract mobile capital and workers (Bartik 1996).¹ Since the early 1980s, a majority of U.S. states have implemented Enterprise Zone (EZ) programs that provide tax credits for investment and job creation in targeted geographic areas within a state.² These programs are costly, with Peters and Fischer (2002) finding that Enterprise Zone tax

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credits were valued around \$5,000 per job for the twenty most industrialized states during the 1990s. Existing empirical studies indicate that the impacts of EZ programs range from positive to even negative on a variety of outcome measures, including employment, wages, investment, and housing prices (Wilder and Rubin 1996; Boarnet 2001).³

One significant problem in estimating the effects of EZ programs is that EZ areas are often economically different than non-EZ areas. Since these economic differences likely influence which regions are designated Enterprise Zones, the problem of self-selection of EZ areas will bias resulting program evaluation. Self-selection occurs in two stages of the EZ designation process.

First, eligibility criteria typically require above-average unemployment and/or low income levels to become a qualifying region.⁴ These eligibility criteria are correlated with unfavorable economic conditions, which can influence the entry of businesses and the hiring of workers. Separating out EZ tax credit effects from local economic conditions requires the creation of adequate control and treatment groups.

Second, an eligible EZ area must apply to the state to receive EZ designation. Some states further complicate the process by requiring applicants to create an economic development plan (including Colorado). The self-selection that occurs at this stage of EZ designation is because of the role of local institutions or community leaders that organize and complete the applications. Also, these institutions and the community leaders themselves influence local economic conditions and subsequent growth rates of EZ areas. This community effect likely varies over time as the application process is completed, designation is determined, and businesses obtain information about the availability of tax credits. This type of self-selection lacks quantifiable measures, which limits most empirical analysis.

This article introduces a methodology of program evaluation that will control for unobservables that may influence the self-selection of EZ regions. In particular, this article uses establishment-level microdata and digitally coded EZ boundaries to match a sample of establishments in close proximity but on opposite sides of the EZ policy border.⁵ Border matching, along with controls for land use, isolates the difference between the EZ and non-EZ areas by the availability of tax credits. Matched establishments are then measured with respect to employment and business location outcomes in EZ areas. These outcome measures are based on Colorado's EZ tax credits, which "provide incentives for private enterprise to expand, for new businesses to locate in economically depressed areas and to provide more job opportunities for residents of such areas" (Enterprise Zone Annual Report 1998).

Results find that EZ tax credits in Colorado insignificantly impact an establishment's location decision. They do however, generate between 1.5 and 1.8 more employees for new establishments locating within an EZ and 0.0 to 0.3 more employees for existing establishments located within an EZ. When delineating new establishments by industry, results find mixed impacts across industries, accentuating the limited role that EZ tax credits have on location decisions. Results also pinpoint the presence of factor specific impacts with a few one-digit Standard Industrial Classification (SIC) industries (agriculture, construction, and retail) experiencing different impacts for employment and location trends in response to EZ tax credits.

This study provides two significant contributions beyond previous research. First, a border-matching estimation strategy controls for observable and unobservable factors that influence the self-selection of EZ regions. Second, specific industry results provide a disaggregated assessment of EZs across all industries, which allows an analysis of industry-targeted incentives within Colorado's EZ program. This article begins with a discussion of relevant literature in section 2 and provides details of Colorado's EZ program in section 3. Section 4 provides a discussion of the data and empirical methodology. Sections 5 and 6 provide impacts for Colorado's EZ program, and section 7 breaks down analysis for specific industry effects.

2. Related Literature

One of the first econometric studies that controls for the endogeneity of EZ designation is Papke's (1994) analysis of the state of Indiana's EZ program. Using a panel of local taxing jurisdictions with similar industrial employment in both Enterprise Zones and non-Enterprise Zones, Papke concluded that EZ designation resulted in an 8 percent increase in the value of inventories and that there was a 19 percent reduction in unemployment claims.⁶ Boarnet and Bogart (1996) found that EZ designation had no significant effect on employment or property values in New Jersey by using an experimental approach in which the municipalities designated EZs were the treatment group and those municipalities that qualified, but were not designated EZs, served as the control group. Engberg and Greenbaum (1999) estimate the capitalization of EZ tax credits in the local housing market, and find that Enterprise Zones have no impact of housing prices. In an examination of five different state EZ programs, Bondonio and Engberg (2000) find no impact on employment and results are robust

to variations in the characteristics of the programs in terms of monetary amounts or program characteristics.

O'Keefe's (2004) study of the California EZ program addresses aggregation biases in past studies and allows for propensity score matching by census tract through the use of establishment-level data. She finds that EZs raise employment by 3 percent each year during the first six years after designation. Greenbaum and Engberg (2004) further contribute to the literature with an analysis of urban EZs in six states. This article uses longitudinally linked manufacturing data which allows more detailed information on EZ programs than previous studies. Propensity score matching and difference-in-difference estimation was used to base comparisons between non-EZ and EZ zip codes and found that EZ programs had a positive effect on the birth of new establishments, but a negative effect on retaining establishments. The absence of consensus among these studies of EZ programs can be attributed to variations in methodologies and EZ program characteristics.

A few articles provide examples of incorporating a border effects methodology for policy and program evaluation. Holmes's (1998) study of state right-to-work laws incorporates the idea that any unobservable characteristic is unlikely to vary between businesses in a small geographical area.

Using the configuration of states with right-to-work laws, he then examined the differences in employment growth between bordering counties that differ in state right-to-work laws. Bronars and Lott (1998) examined concealed weapon laws using differences across state borders and found that "shall issue" concealed gun laws create geographic spillovers in crime to neighboring counties without such laws. Finally, Holcombe and Lacombe (2004) use a "matched" border technique to find that Aid to Families with Dependant Children (AFDC) and food stamp programs have a positive impact on the number of female-headed households and a negative effect on female labor-force participation. In these border methodology studies, the data used are at a larger scale than this study of Colorado's EZ program. S. Black (1999) provides a comparable methodological approach to value school quality. Her analysis at the border of school attendance zones finds a positive impact of primary school test scores on housing prices.

3. Colorado's Enterprise Zone Program

Colorado's Enterprise Zone Program began in 1986 and currently limits the number of distinct enterprise zones to 16.⁷ Fourteen of the sixteen

zones were formed by 1990, the Jefferson County EZ formed in 1991, and the Larimer County EZ formed in 1993.⁸ The sixteen EZs vary in size from urban EZs of less than a square mile to rural, multicounty EZs containing over one thousand square miles. Results from analyzing EZs in Colorado are overwhelmingly driven by urban EZs, and the border matching methodology limits conducting a separate analysis of sparsely developed rural EZs. Overall, 22.8 percent of establishments that existed between 1990 and 2000 were located within an EZ.⁹ Eligibility for EZ designation requires the area to have a population of less than eighty thousand and meet at least one of the following criteria:

- (1) Unemployment rate at least 25 percent above state average.
- (2) Population growth rate less than 25 percent of the state average.
- (3) Per capita income less than 75 percent of the state average.¹⁰

An eligible jurisdiction must then create an economic development plan and apply for EZ designation with the state. EZ boundaries include some combination of census or political geographies (e.g., census tracts, zip codes, etc.) to demonstrate eligibility using readily available census or state data.¹¹ An establishment can locate, increase employment, or provide employee benefits within an EZ to receive a number of different tax credits.¹² Some tax credits are linked to job creation and investment in specific industries and/or activities including agriculture, manufacturing, mining, and research and development. A 3 percent investment tax credit is offered to businesses making investment in equipment used exclusively in an EZ. This investment tax credit accounted for nearly 70 percent of all EZ tax credits (Enterprise Zone Annual Report 1996). The second most commonly used tax credit is that of the \$500 per job created, which is given to new establishment jobs or for employment expansion by existing establishments (Enterprise Zone Annual Report 1996).¹³ The total amount of tax credits given is substantial, as Colorado averaged \$30 million a year in foregone tax revenue because of the EZ program between 1986 and 1996. The amount has increased to over \$60 million annually between 2000 and 2005 (Enterprise Zone Annual Report 2005).

A previous analysis of the Colorado EZ program was done by Alm and Hart (1998). Using 1980 and 1990 census data, they concluded that the program has positive and significant impacts on both employment growth and per capita income. Lynch and Zax's (2004) study of Colorado's EZ program finds EZ status had a positive impact of 9.8 employees for each establishment that existed in 1990 and survived until 2000 using a Tobit

model. The incorporation of a Heckman selection model produced reverse results by showing a negative impact of 19.7 employees.

4. Method

To properly measure Colorado's EZ tax credits targeted for investment and job creation, one has to model how different tax credits will influence an establishment's profit-maximizing choices of location, investment, and employment. Since EZ tax credits are primarily targeted to new establishments in Colorado, modeling their behavior is based on two decisions. An establishment chooses a location for its business and its corresponding level of capital and labor. The availability of tax credits should encourage businesses to locate in EZ regions and also lower the costs of expanding existing businesses. The effects on entry are a result of business owners recognizing the value of EZ tax credits, and new businesses forming that would not expect profitability outside the EZ. The amount of job creation from new businesses will depend on industry characteristics, as well as EZ incentives for employment. The effects of EZ tax credits on entry and employment will provide a measurement of the overall effectiveness of the EZ program in Colorado.¹⁴

As discussed, a number of methods have been employed in past research to control for the self-selection of EZ regions. The use of propensity score-matching models and difference-in-difference techniques controls for observable factors and existing EZ region trends that can influence EZ designation. One issue not controlled for in these techniques is the presence of unobservable factors, which may vary over time and are correlated with EZ designation and the entry and/or employment decisions of an establishment. A difference-in-difference methodology may control for some unobservable confounding variables, but this technique has an underlying assumption that unobservable factors are time invariant (Cameron and Trivedi 2005). Intuitively, factors that influence EZ designation affect the growth rates of outcome variables in an EZ area versus non-EZ area. Greenbaum and Engberg (2004) discuss the potential problem that EZs may have fundamentally different growth trends, which requires a time-variant technique to control for self-selection factors. Since the change in growth rates results in different time trends of outcome measures, this makes difference-in-difference estimation dependent on time trends that are correlated with EZ designation. Resulting estimates will therefore depend on which time intervals are used for differencing.

A border-matching methodology that matches only EZ and non-EZ areas in close geographical proximity will control for time-varying unobservables by limiting comparisons between EZ and non-EZ outcomes to neighboring commercial areas. The idea that locations close together are more alike is well established in the geography literature as Tobler's Law and forms the basis for spatial econometric models. Extrapolating this idea to EZ regions is straightforward because factors such as crime, access to markets, and agglomeration economies are likely to be the same within a small area. Therefore, a policy border allows a segmentation of commercial areas into control and treatment groups. The control group is represented by the area just outside the EZ policy border, and the treatment group is the area just inside the EZ policy border. Since EZ tax credits focus on new businesses and jobs, an establishment's location decision will take existing economic conditions and the EZ policy border as exogenous. Therefore, the estimated impacts of EZ tax credits is represented by the differences in business location trends or employment between the non-EZ and EZ areas.¹⁵

4.1 Exogeneity of Border Location

Border matching assumes an exogenously determined policy border. If factors that influence the border of the EZ also influence outcome measures, then this assumption is invalid. The EZ designation process limits the location of the EZ border in two manners. First, state legislation limits the population contained within an EZ, which limits the size and expansion of an EZ area. Second, EZ boundaries are limited to census or political geographies in determining eligibility based on unemployment, income, or population growth measures. Other factors related to neighborhood characteristics that may influence EZ designation are unlikely to vary significantly in such a small geographic area. The main factor that is endogenous to an EZ's border location is commercial land use and zoning limitations. The incorporation of zoning data is problematic because of the small geography of interest and the dynamic nature of zoning and land use. A reasonable proxy for land use is the location of existing establishments. Empirical analysis incorporates two measures of commercial land use to control for zoning limitations. Results report the impacts of EZ tax credits for only border areas with similar existing 1990 number of establishments within and outside the EZ. The second measure of land use only includes border areas with similar numbers of previous year deaths of existing establishments. Since the death of an establishment in the previous year is exogenous to tax

credits (which are based only on newly created establishments and jobs), this measure represents the number of available commercial properties for new establishments to the border area.

4.2 Data Source and Information

Specific information on Colorado's EZ program is from the State of Colorado's Department of Local Affairs. The geographic information obtained from the individual zone coordinators were used to create digital maps of each EZ.¹⁶

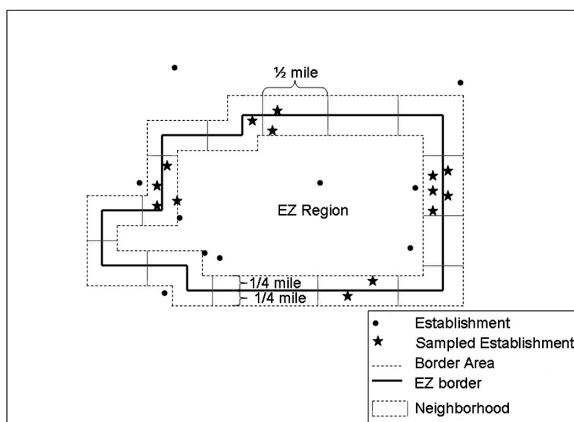
Data on establishments and employment is from the ES-202 database, which contains quarterly data on employment, by industry, at the establishment level.¹⁷ The physical addresses of the establishments obtained from the ES-202 database were geocoded to their geographical locations based on the 2000 TIGER Census Address Database. ESRI ArcGIS 9.0 software was used to determine an establishment's location relative to the digitized map of EZ boundaries. Outcome variables for ES-202 data are establishment employment, total payroll, four-digit SIC, and an indication of an establishment's existence on a quarterly basis. This research aggregated the ES-202 data to a panel of annual averages for 1990 through 2000. The use of annual averages matches the process used by the EZ program to determine employment tax credits.

From the state ES202 database, 282,988 establishments that existed for at least one year in Colorado between 1990 and 2000 were geocoded in ArcGIS 9.0. Seventeen percent of the establishments given in the ES202 records could not be geocoded because of incomplete addresses, missing addresses, or lack of a street address.¹⁸ A concern with losing a portion of the data would occur when establishments located just inside the border of the Enterprise Zone are systematically different from those just outside the Enterprise Zone in their likelihood of providing a physical, geocodable address. This is highly unlikely given that the EZ Program and ES202 records are compiled separately and for different purposes by the state. Nongeocodable establishments were found to have similar measures for all ES202 variables as geocodable establishments.¹⁹

4.3 Empirical Method

The nature of the border-effects methodology as applied to EZs is highlighted in figure 1. The border area between the dotted lines represents the region where the sample of border establishments is taken and is based on

Figure 1
EZ Border Methodology



Note: EZ = Enterprise Zone.

a one-quarter mile buffer around the EZ border.²⁰ The points represent establishments and each establishment in this sample is given an indicator to represent location within or outside the EZ benefit area. The segmented regions of one-half mile along the EZ border represent neighborhoods and are in essence a local fixed effect which maintains comparisons between border establishments in close proximity. The sample of border establishments incorporated in later analysis is represented by the stars in figure 1.

To create EZ border neighborhoods, this research constructed a square lattice of points one-half mile apart across the entire state of Colorado using Geographical Information Systems (GIS).²¹

Connecting these points created one-half-mile-wide squares, which were then overlaid onto digitized EZ boundaries. The lines representing the borders of the squares partitioned the EZ border into segments whose length was within the one-half mile square. A total of 55,952 establishments that existed for at least one quarter during 1990 through 2000 and were within a one-quarter mile of the EZ border were assigned to the closest border segment. This methodology created a total of 3,216 neighborhoods, of which 812 neighborhoods contained at least one establishment for at least one year between 1990 and 2000.²² A total of 481 neighborhoods contained establishments on both sides of the EZ border.

Two metrics will control for land use in later estimation results. The metrics involve matching EZ and non-EZ areas within a border neighborhood based on existing 1990 establishments or previous year establishments lost. The land-use matching criteria includes neighborhoods (j) that satisfy a 20 percent difference in land-use measure (Z) between EZ and non-EZ areas as shown in equation 1.

$$\frac{|Z_j(EZ=1) - Z_j(EZ=0)|}{1/2(Z_j(EZ=1) + Z_j(EZ=0))} \leq 0.2 \quad (1)$$

Matching based on 1990 establishments resulted in 74 border neighborhood observations over 11 years. Matching on previous year deaths included 174 different neighborhoods and a total of 614 observations. Land-use matching excluded neighborhoods with $Z_j(EZ=1)$ or $Z_j(EZ=0)$ equal to zero. Table 1 provides descriptive statistics for ES202 data for census tracts, border neighborhoods, and neighborhoods matched on 1990 existing establishments.²³ The data provided in table 1 allows an examination of how well border matching controls for differences between EZ and non-EZ areas. The progression from all data at the census-tract level to border neighborhoods to land-use matched neighborhoods demonstrates that this methodology is controlling for differences between the EZ and non-EZ area. The number of employees (full- and part-time) per establishment is greater in EZ areas than non-EZ areas. In general, EZ areas contain more commercial activity than non-EZ areas as shown by the average annual number of establishments in border neighborhoods. The difference between average annual number of new establishments and establishments lost among border neighborhoods is 0.48 establishments for EZ areas and 0.5 for non-EZ areas. Land-use controls remove any differences in commercial activity and overall trends for establishment location are similar between the EZ and non-EZ areas. The fraction of establishments that are part of a multiestablishment firm is larger in border EZ areas, and estimation will control for this attribute. The fraction of establishments that are part of a multiestablishment firm provide a measure of an establishment's organizational structure and may proxy for any informational differences between single and multiestablishment firms on the availability of EZ tax credits.

Table 1 reveals some differences in industrial composition between the EZ and non-EZ border areas. There is a larger representation of manufacturing, wholesale trade, and retail, and a smaller representation of construction and services in EZ areas. The results for agriculture and manufacturing could be because of the presence of additional EZ tax credits targeted to these industries.

Table 1
Descriptive Statistics

	Census Tracts		Border Neighborhoods		Land-Use Matched Border Neighborhoods	
	<i>In EZ</i>	<i>Not in EZ</i>	<i>In EZ</i>	<i>Not in EZ</i>	<i>In EZ</i>	<i>Not in EZ</i>
Annual Means (1990–2000) (Standard Deviation)						
Employees (full and part-time) per establishment	15.7 (51.9)	13.5 (43.4)	13.9 (21.1)	9.7 (11.1)	23.1 (48.2)	21.2 (52.1)
Fraction of establishments that are part of a multiestablishment firm	0.15 (0.18)	0.15 (0.19)	0.11 (0.16)	0.08 (0.14)	0.15 (0.15)	0.13 (0.11)
Number of establishments in a neighborhood	79.8 (183.0)	20.1 (126.7)	32.4 (89.1)	21.2 (82.2)	44.2 (123.2)	41.1 (133.2)
Number of new establishments in a neighborhood	11.8 (25.5)	4.0 (23.1)	4.3 (11.7)	3.3 (12.5)	6.8 (18.8)	6.8 (22.1)
Number of establishments lost in a neighborhood	10.4 (23.2)	3.1 (19.0)	3.9 (10.8)	2.8 (10.8)	5.9 (17.6)	5.8 (20.0)
<i>Industry Composition for One-digit SIC by Census Tract/Border Neighborhood</i>						
Agriculture	4.8%	3.7%	7.5%	5.0%	5.4%	5.6%
Mining	1.9%	1.3%	2.1%	2.1%	1.6%	1.4%
Construction	13.2%	8.6%	21.9%	26.5%	16.2%	18.3%
Manufacturing	11.7%	8.5%	6.4%	4.0%	7.3%	4.9%
Transportation, communication, and utilities	4.1%	3.4%	6.4%	6.4%	6.5%	5.9%

(continued)

Table 1 (continued)

Annual Means (1990–2000) (Standard Deviation)	Census Tracts		Border Neighborhoods		Land-Use Matched Border Neighborhoods	
	<i>In EZ</i>	<i>Not in EZ</i>	<i>In EZ</i>	<i>Not in EZ</i>	<i>In EZ</i>	<i>Not in EZ</i>
Wholesale trade	3.2%	3.5%	7.3%	5.9%	8.4%	6.8%
Retail trade	19.1%	28.8%	16.9%	12.6%	19.7%	16.9%
Finance, insurance, and real estate	10.4%	8.7%	4.6%	5.5%	6.6%	6.9%
Services	31.0%	33.3%	26.4%	31.0%	24.3%	28.4%
Government	1.6%	1.5%	1.1%	0.9%	0.7%	0.5%
Number of observations	1,122	3,707	4,810	4,810	814	814

Note: EZ = Enterprise Zone. Descriptive statistics are only based on 2000 census tracts/border neighborhoods with positive establishment counts. Land-use matched neighborhoods are based on similar number of 1990 existing establishments.

5. Results for Entry Decision

This section considers the impacts of EZ tax credits on an establishment's entry decision. Examining the entry of establishments provides an impact of both job creation and investment tax credits. Corresponding measures of exit or establishments lost will reveal the longer-term impacts of tax-credit induced entry. New establishment location choices and the corresponding choices of labor and capital can result in different impacts between the entry and factor of production decisions. This is partly attributed to factor specific tax credits for job creation and investment, which can alter relative factor prices between capital and labor. If the cross price elasticity between capital and labor is positive, then EZ tax credits encourage factor substitution and decrease EZ impacts on employment or investment relative to non-EZ areas. Measurement of establishment location trends and later analysis of employees per establishments will highlight respective impacts on entry and the subsequent tax-credit induced use of labor.

Table 2 provides analysis of impacts of EZ status on the number of new establishments and number of establishments lost between 1990 and 2000. The basis for this analysis is that an establishment first decides to locate in a given commercial area based on the set of location characteristics available. For those establishments locating in a border neighborhood, the establishment then must choose between sites with an EZ tax credit and ones without EZ tax credits. Significant differences in establishments choosing EZ locations will highlight the impact of the tax credits. The scale of analysis for the location of establishments is based on border neighborhoods, which allows for a comparison of similar square mileage of land on each side of the EZ border. Mean values of number of new establishments and number of establishments lost are compared between EZ and non-EZ border areas. The reported results give the difference in means between the EZ and non-EZ area. The total number of observations is given in table 2 and bootstrapped standard errors are incorporated for all hypothesis tests on differences in means and are reported in parentheses.

Table 2 adopts nonparametric differences in means specifications for border matching as well as propensity score matching. A propensity score matching model based on census tracts is provided to compare results with previous EZ studies and as a method to provide EZ impacts without controlling for time-varying unobservables. A nearest neighbor matching based on a propensity score logit regression determines matched census tracts.²⁴ Only matched census tracts with propensity score differences of

Table 2
Border Establishment Location Trends (1990–2000)

Difference in Annual Means (EZ minus non-EZ Areas)	Number of New Establishments		Number of Establishments Lost	
	1990–1995	1996–2000	1990–1995	1996–2000
Census Tracts				
All census tracts ($n = 4,829$)	-0.24 (1.19)	-2.11* (1.28)	0.96 (1.11)	-0.61 (1.18)
Propensity score matched ($n = 1,082$)	4.31 (3.10)	4.18 (3.04)	4.84** (2.31)	3.82 (3.01)
Border Neighborhoods				
All border neighborhoods ($n = 4,810$)	0.90*** (0.30)	1.08*** (0.41)	0.90*** (0.26)	1.11*** (0.35)
Matched on 1990 establishments ¹ ($n = 814$)	1.50 (2.60)	2.46 (3.42)	0.68 (2.30)	2.11 (3.16)
Matched on death ($t - 1$) ¹ ($n = 614$)	-0.43 (4.04)	0.61 (3.84)	-0.16 (3.54)	1.09 (3.43)

1. Includes only neighborhoods where EZ and non-EZ areas have less than 20% difference in the matched variable for land use.

Note: n represents the number of observations for the entire panel 1990–2000. Standard errors are bootstrapped and in parentheses.

Significant at *.010; **.005; ***0.01.

less than 10 percent are included in results. O’Keefe (2004) provides a good explanation of this methodology in the context of EZs.²⁵

Results indicate that border areas matched on land use have an insignificant impact of EZ status on the number of new establishments and establishments lost annually. The robustness of this result across specifications that control for land use highlights that EZ tax credits have limited or no influence on establishment location trends. Results were consistent between the first half (1990–1995) and the second half of the panel (1996–2000). This highlights that the initial time period after which most EZs were formed (late 1980s) did not create differential impacts. This is consistent with the tax credits in Colorado’s EZ program, which are available to new investment and job creation and have not changed over the 1990s. Results are robust to the use of 0.1, 0.3, and 0.4 for land-use matching thresholds in equation 1. Also, results did not change when examining only multiestablishment firms.

The propensity score model provides significant results for establishments lost in 1990 through 1995. Tax credits should not influence establishment

exit because they are given only for entry and job creation. One explanation is that tax credits may induce entry for less competitive establishments, which are more likely to exit in subsequent years. Tax-credit induced entry is only plausible if new establishments do not anticipate their inability to be competitive businesses, and tax credits influence the formation of new establishments. The formation of an establishment because of a tax credit is a strong and unlikely assumption. This research adopts the weaker assumption that tax credits only influence the location decisions of an establishment at the neighborhood level.

6. Results for Job Creation

Multivariate analysis tests the impact of EZ status on job creation at the establishment level.²⁶ This impact is caused by a variety of elements of the EZ tax credit program. The job creation tax credit and the employment training tax credit both directly benefit establishments that employ more workers. Also, job creation occurs when larger establishments benefit from investment and/or job creation tax credits and therefore create more jobs within the EZ. Since employment levels likely vary by industries and ownership structure, regression analysis includes a multiestablishment dummy variable and fixed effects for year and two-digit SIC. Fixed effects for each border neighborhood will limit border comparisons to small geographic areas. Table 3 highlights impacts on existing establishments and impacts on new establishments in separate regressions.

The regressions in table 3 are sparse in explanatory variables because the border methodology controls for other factors.²⁷ The top set of regressions in table 3 indicates results for establishments that existed in time t and time $t - 1$. The bottom set of regressions in table 3 provides results for new establishments in time t . A number of border matching specifications, including a comparison to a Propensity Score (PS) matching model, provide EZ impacts. The coefficient on the EZ indicator variable represents the difference in employment changes for establishments within an EZ versus those just outside the EZ. Similar interpretation holds for the multiestablishment firm indicator variable.²⁸

All specifications except for existing establishments with land-use controls provide positive results. The coefficient on the indicator variable for EZ status ranges from 0.0 to 0.3 more employment for existing establishments and from 1.5 to 1.8 greater employment for new establishments. The differences between the border matching and propensity score results

Table 3
Enterprise Zone Impacts on Employment

Employment Growth for Existing Establishments			
	Census Tracts <i>Propensity Score</i>	Border Neighborhoods	Border Neighborhoods <i>Match on 1990 Number of Establishments</i>
Dependant Variable = $\text{Emp}(t) - \text{Emp}(t - 1)$ (For establishments with prior year employment)			Border Neighborhoods <i>Match on death (t - 1)</i>
Enterprise Zone	0.33*** (0.12)	0.28*** (0.10)	0.29 (0.21)
Multiestablishment dummy	1.47*** (0.19)	1.50*** (0.20)	2.07*** (0.52)
Neighborhood/census tract fixed effects	Yes	Yes	Yes
Two-digit SIC dummy variables	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes
R-squared	0.03	0.14	0.17
Observations	285,317	160,354	42,393
			51,721

Table 3 (continued)

Employment Growth for New Establishments			
Dependant Variable = $\text{Emp}(t) - \text{Emp}(t - 1)$ (For establishments with prior year employment)	Census Tracts <i>Propensity Score</i>	Border Neighborhoods	Border Neighborhoods <i>Match on 1990 Number of Establishments</i> <i>Match on death (t - 1)</i>
Enterprise Zone	1.80** (0.90)	1.75** (0.83)	1.51** (0.74)
Multiestablishment dummy	20.25*** (1.63)	23.54*** (2.08)	32.53*** (4.99)
Neighborhood/census tract fixed effects	Yes	Yes	Yes
Two-digit SIC dummy variables	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes
R-squared	0.19	0.14	0.26
Observations	76,927	31,894	9,187

Note: Absolute value of standard deviation in parentheses. All regressions include clustered standard errors by establishments.
* significant at 10%; ** significant at 5%; *** significant at 1%.

highlight that PS matching produces larger estimates than border matching. This is consistent with a concern that unobservables, which are likely positively correlated with EZ designation and establishment outcome variables, may positively bias the estimated impacts of EZ tax credits.

Overall, the smaller impacts for existing establishments is not surprising because EZ tax credits only apply to establishments that expand employment by at least 10 percent or investment by at least \$1 million. The larger impacts for new establishments are consistent with the nature of the EZ tax credits, which are available to all new establishments. Also, the ability of a new establishment to initially determine levels of employment provides greater flexibility in modifying factor inputs in response to the tax credit.

7. Industry Results

Often local economic development programs target specific industries that have desirable characteristics for job creation and wages. This has lead to the focus of a number of regional development efforts on large manufacturing facilities (see Greenstone and Moretti [2004] for an article on how local policy is used to attract large manufacturing companies) or high technology industries to foster the development of industry-based clusters (based on the model of industry clusters popularized by Porter [1990]). Targeted industry incentives in Colorado's EZ program are represented by additional tax credits for agriculture, manufacturing, mining, and research and development activities. The presence of industry level impacts, even when overall impacts are insignificant, may signify the need to better focus EZ tax credits on the industries that have demonstrated response to specific incentives.

7.1 Entry Decision

Industry results for new establishment entry emphasize the impacts of job creation and investment tax credits. Table 4 provides border location trends for new establishments by one-digit SIC industry. Analysis for establishments lost was not possible given the low number of border neighborhoods with existing establishments for a given one-digit SIC on both sides of the EZ border. Three different specifications are provided in table 4 for the difference in annual number of new establishments between the EZ and non-EZ border neighborhoods over the 1990 through 2000 time period.

Table 4
New Establishment Location Trends by SIC (1990–2000)

Difference in Annual Means (EZ minus non-EZ Areas)	Number of New Establishments		
	Border Neighborhoods	Border Neighborhoods Match on 1990 Establishments	Border Neighborhoods Match on Death ($t - 1$)
Agriculture	0.0003 (0.01)	−0.03 (0.03)	0.04 (0.04)
Mining	−0.17 (0.12)	−0.26* (0.14)	−0.34* (0.19)
Construction	−0.039** (0.02)	−0.39** (0.11)	−0.41*** (0.14)
Manufacturing	0.16*** (0.01)	0.16*** (0.06)	0.30*** (0.08)
Transportation, communications, utilities (TCU)	0.05*** (0.01)	−0.15 (0.09)	−0.34 (0.13)
Wholesale	0.16*** (0.02)	−0.21* (0.12)	−0.01 (0.16)
Retail	0.32*** (0.04)	−0.03 (0.25)	0.77** (0.32)
Finance, insurance, real estate	0.009 (0.04)	−0.71* (0.43)	−0.82 (0.55)
Services	0.32*** (0.11)	−0.24 (1.07)	0.25 (1.49)
Government	0.0002 (0.08)	−0.06 (0.07)	−0.05 (0.11)
<i>N</i>	4,810	814	614

Note: Land-use matching based on Enterprise Zone (EZ), non-EZ areas containing less than 20% difference in the matched variable for land use. Standard errors are bootstrapped and in parentheses; *n* represents the number of observations for the entire panel 1990–2000.

In general, results represent significant impacts of EZ tax credits on the location decisions of new establishments in some industries even though overall location trends for new establishments are insignificant. Positive results are found for manufacturing and retail, and negative results occur for mining and construction. An additional EZ tax credit for manufacturing investment is consistent with greater entry by new establishments in manufacturing. The presence of insignificant and negative results for some industries is attributed to two factors. First, tax credit benefits may be capitalized in land rents, which limits the benefits for new establishments.

Second, some industries may lack a set of acceptable locations on both sides of the EZ border because of site/parcel requirements and/or natural resource access, which may result in negative impacts inside the EZ. Mining and construction likely have a restricted set of acceptable locations because of zoning, land availability, and access to infrastructure or natural resources.

The limited or inconsistent influence of EZ tax credits on location decisions may be explained by the fact that landowners are able to capitalize these tax credits into rents, negating the net benefit to businesses.²⁹ The capitalization of EZ tax credits should not impact job creation because greater labor intensity positions establishments to incur more land rent than lower labor intensity establishments. The benefit of the tax credit will be realized by the landowner, but job creation will still occur. The insignificant impacts of establishment location could also signify the small impact of EZ tax credits compared to other site or parcel characteristics. The literature highlights the possibility of both explanations, where McDonald (1993), Man (1995), and Wheaton (1984) find partial capitalization of commercial property taxes into commercial rents. Erickson and Syms's (1986) study of British enterprise zones finds that landlords captured over 60 percent of tax incentives in higher rents.

7.2 Job Creation Analysis

Table 5 extends earlier regressions to include interaction terms between the EZ indicator variable and a dummy variable for an establishment's one-digit SIC. As before, analysis is broken down into existing and new employment. Agriculture, construction, manufacturing, wholesale, and service industries report positive impacts for EZ status. The results for manufacturing are consistent with the entry of new manufacturing establishments in EZ areas. An additional tax credit for agriculture job creation supports the positive results for the agriculture sector.

The results for agriculture, construction, and retail employment impacts differ from new establishment location results for these industries. Agriculture industries had insignificant results for new location trends and construction had negative results. Retail had positive impacts on entry and had insignificant employment growth. These results for agriculture, construction, and retail could be a result of factor substitutability between labor and capital intensive production in response to factor-specific EZ tax credits. The possibility of factor substitutability is a concern for the type of tax credits offered in an EZ program. Overall conclusions on the

Table 5
Enterprise Zone Impacts on Employment by One-Digit SIC

Dependant Variable = Employment(<i>t</i>) – Employment(<i>t</i> – 1)	Existing Establishments		New Establishments	
	<i>No Land-Use Match</i>	<i>Match on 1990 Establishments</i>	<i>No Land-Use Match</i>	<i>Match on 1990 Establishments</i>
Multiestablishment Dummy	1.51** (0.20)	1.87*** (0.43)	23.60** (2.09)	30.08*** (3.84)
Agriculture Dummy*Inez	0.64** (0.24)	0.58* (0.33)	1.24* (0.65)	1.08* (0.59)
Mining Dummy*Inez	0.636* (0.38)	0.686* (0.39)	2.9 (3.13)	1.55 (3.92)
Construction Dummy*Inez	1.15*** (0.23)	0.71* (0.40)	2.31*** (0.70)	4.24 (6.40)
Manufacturing Dummy*Inez	0.47 (0.38)	0.50* (0.28)	11.54*** (4.39)	14.70** (7.35)
TCU Dummy*Inez	–0.53 (0.80)	–0.98 (1.84)	–13.4 (10.48)	–9.83 (17.60)
Wholesale Dummy*Inez	0.02 (0.14)	0.05 (0.23)	2.33** (1.01)	1.67** (0.80)
Retail Dummy*Inez	–0.10 (0.11)	–0.16 (0.13)	–1.61 (1.31)	–0.98 (1.23)
FIRE Dummy*Inez	0.16 (0.16)	0.40 (0.31)	0.36 (0.95)	–0.08 (1.96)
Service Dummy*Inez	0.37** (0.18)	0.26 (0.34)	1.61 (1.31)	4.99* (2.97)
Government Dummy*Inez	1.45 (0.84)	–0.54 (0.83)	8.95 (6.11)	10.02 (13.74)
One-half mile border segments fixed effects	Yes	Yes	Yes	Yes
Two-digit SIC dummy variables	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes
R-squared	0.14	0.17	0.14	0.26
Observations	160,354	42,393	31,894	7,770

Note: TCU = transportation, communications, utilities. Absolute value of standard deviation in parentheses. All regressions for existing establishments include clustered standard errors. Land-use matching based on number of 1990 establishments.

presence of factor substitutability in Colorado's EZ program are limited by the unavailability of data on an establishment's capital intensity.

Establishments may also perceive the value of the job-creation tax credits in Colorado as too small to impact overall employment decisions. Bondonio and Engberg (2000) find that the amount of the fiscal incentive does not influence results for five different state EZ programs, while O'Keefe's (2004) results find that a decrease in the magnitude of EZ tax

credits over time corresponds with a decrease in estimated impacts on employment growth. These inconclusive results from the literature preclude reasonably predicting the effects of changing the magnitude of job creation tax credits in Colorado. Given controls for other factors that may influence labor intensity, even a small tax credit should influence the marginal employment decision. Results from this article weakly support the idea that larger job-creation tax credits positively influence labor intensity and job creation.³⁰ The overall positive impact of EZ status on job creation for new establishments indicates EZs attract larger establishments.³¹ Since larger establishments are eligible for a greater total amount of tax credits, this would support the hypothesis that larger tax credits can positively influence job creation. Agricultural industries are eligible for a double job-creation tax credit and they experienced positive impacts of EZ status on job creation. Manufacturing establishments experience positive impacts on entry and are eligible for larger investment tax credits.

8. Conclusion

The influence of policy on the location of economic activity is just beginning to be understood in the literature, and a number of place-based government programs have been the focus of past analysis. This research provides a new empirical methodology for examining the impact of tax incentives on the location of jobs and businesses and tests it on a commonly used tax-incentive program in the United States, the Enterprise Zone program. The problems of spatial mismatch between EZ and data geography and the presence of unobservables have been problematic in a number of earlier works in this area. The use of border effects and establishment-level data takes advantage of the spatial nature of these programs to overcome these problems, which provides cleaner estimates of EZ impacts in Colorado. Overall, results find insignificant impacts from the EZ program on the location decision of establishments, but positive impacts for job creation. Industry results highlight the importance of taking specific tax credits into consideration in examining factor specific outcomes. Results have limited generalization beyond Colorado given that states vary in how EZs are designated and the monetary amounts of EZ tax credits. Overall results do highlight that evaluating the impact of tax incentives needs to account for time-varying factors that may influence the use and outcome of tax incentives.³²

Notes

1. As discussed by Buss (2001), even with the billions of dollars spent annually, there are a limited number of rigorous studies of tax incentive programs.

2. According to Peters and Fischer (2002), approximately forty states as well as the District of Columbia have implemented an EZ program in one form or another. While these programs vary in specifics, they are all aimed at stimulating economic development in depressed areas.

3. Given that tax credits should only benefit businesses, the finding of negative impacts highlight the difficulty in controlling for other factors that influence business location and hiring decisions.

4. High unemployment, low income, and high poverty rates relative to a state's average are used as part of eligibility criteria for most states (Peters and Fischer 2002).

5. Of note is the terminology used throughout this article: establishments are single, physical locations, while a firm is the legal entity through which an establishment is formed. Establishments are subsets of firms.

6. This methodology represents propensity score matching based on industrial composition. For EZs, this technique involves creating a control and treatment set of EZ regions based on characteristics such as industrial composition, income, population, and matching similar regions which only differ in EZ status. The matched border technique used in this article represents a form of PS matching based on location relative to the EZ border.

7. This is based on existing legislation in 2000.

8. Given that the data in this analysis is limited to 1990 through 2000, the Jefferson and Larimer County EZs were examined separately with pre- and post-EZ designation observations, and the number of observations were too limited for any conclusions.

9. Based on author's calculations.

10. These eligibility requirements are based on Colorado State Statute: Urban and Rural Enterprise Zone Act (Title 39, Article 30).

11. In a few cases, EZ boundaries are designated at a subcensus or political geography. In these cases, EZ boundaries are created around only commercial land use and designation is based on the larger geographies' characteristics. These cases are almost all excluded from border analysis because of the lack of observations outside the EZ. Results are unchanged when these specific EZs are excluded.

12. A complete list of all available tax credits is published on the state of Colorado's Office of Economic Development and International Trade website, <http://www.state.co.us/oed/enterprise-zone/EZ-Tax-Credits.cfm>.

13. The Colorado EZ Program formula for the job creation tax credits is given by, $\text{EZ Job Creation} = (\text{Average Annual Employment in year } t) - (\text{Average Annual Employment in base year})$. The base year of employment is determined by the highest employment level from any prior annual employment level. The base year is set equal to zero for any new establishment.

14. The unavailability of data on capital limits directly measuring impacts on capital intensity.

15. As noted by an anonymous referee, EZ tax credits may alter the growth path of EZ areas. Growth path impacts in this research are limited to those impacts that accrue to only the EZ treatment group and not the matched control group.

16. Digital maps were created by Devon Lynch (University of Denver) based on paper maps from each individual EZ. During the 1990 through 2000 time period of study, EZ status was not removed even if economic conditions improved.

17. This database is compiled as a cooperative endeavor of the Bureau of Labor Statistics (BLS) and the employment security agency of the State of Colorado.

18. The portion of ES202 data that was geocodable is comparable to that of O'Keefe's (2004) study.

19. One plausible case that could result in problems is if multiestablishment firms are more likely to give a nonphysical location address or a corporate headquarter address for their ES202 records, and are less likely to respond to EZ incentives. In this case, estimates of EZ effects will be biased because of the underrepresentation of certain types of establishments that may have experienced higher or lower economic growth over this time period. To control for this potential problem, estimation accounts for multiestablishment firms and SIC industries.

20. The choice of the one-quarter-mile buffer is based on the need to limit the distances between matched establishments and the spatial configuration of EZ boundaries. The presence of differences in agglomeration benefits, market area, or natural resource advantages is unlikely to differ at this scale (see Rosenthal and Strange [2001] for a discussion on the scale of agglomeration benefits). Overall results are robust to one-half-mile border buffers and are available from the author on request.

21. The initial point of the lattice is based on the northwest corner of the state.

22. The large number of neighborhoods not assigned any establishments is because of a number of neighborhoods created along rural EZ borders and residential areas that contain no commercial or limited commercial land use.

23. Given that the border-effects methodology does not follow census geographies, census-based summary data cannot be included in table 1.

24. This logit regression includes 1980 census variables for population, income, employment, ethnicity, and housing and 1990 ES202 data for number of establishments. Estimated coefficients are available from the author on request.

25. See Black and Smith (2004) for a discussion of propensity score matching models and the problems of self-selection in estimating treatment effects.

26. The unavailability of establishment-level data for measures of capital prevents directly testing the impact of investment tax credits.

27. Because of concerns of heteroscedasticity and the fact that this is a panel dataset, all regressions on existing employment incorporate clustered standard errors at the establishment level within a robust variance correction. This clustered variance correction is based on White (1980) and Woodbridge (2002).

28. Unreported regressions with an interaction term between the EZ dummy and multies-establishment dummy were insignificant for border matching specifications in table 3.

29. Lynch and Zax (2004) discuss the potential capitalization on EZ tax credits into land rents.

30. This idea is also consistent with de Bartolome and Spiegel's (1997) finding that higher economic development spending at the state level is positively correlated with manufacturing employment.

31. The size of an establishment in this context is based on employment.

32. A generalization for the border-matching methodology is possible by examining the growth of establishments or employment from before to after the designation of an EZ and comparing only establishments near the border of the newly created EZ. Unfortunately, the limitation of earlier years of ES202 data prevents this methodology from being incorporated into this research, but may be feasible in other states.

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